

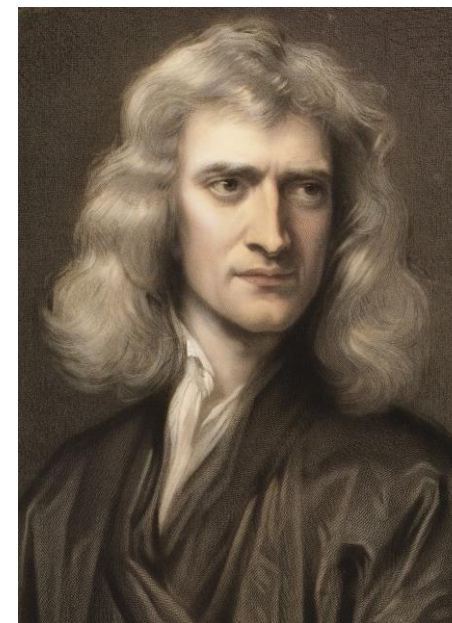
Math Primer

(10. Partial Derivatives)

Contents



- ▶ Functions of several variables
- ▶ First-order partial derivatives
- ▶ Second order and mixed partials
- ▶ Total Derivative
- ▶ Multivariate Taylor Series
- ▶ Chain Rule
- ▶ Gradient
- ▶ Jacobian
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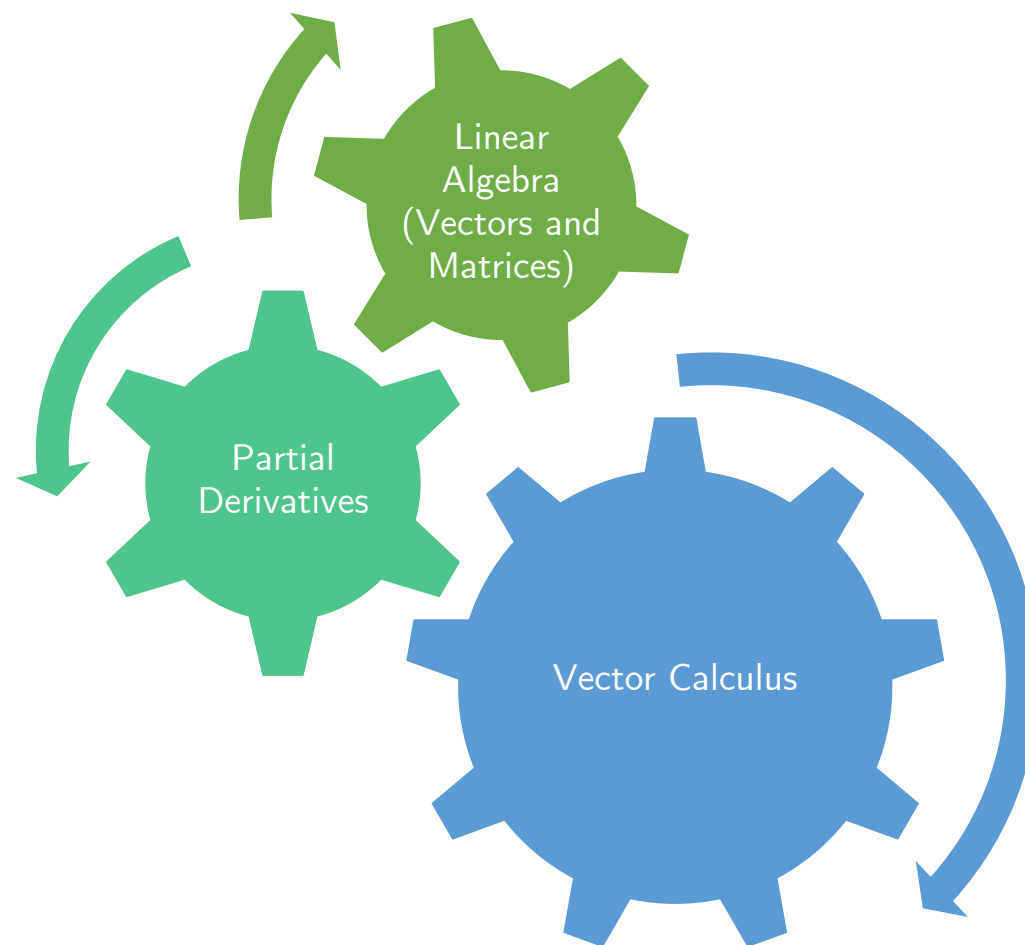


Isaac Newton
(1643-1727)



G.W. Leibniz
(1646-1716)

The Multivariable Calculus Landscape





Functions of several variables

- Often, we come across functions which depend on multiple input variables. We call those multivariable functions (function of several variables). A bivariate function for example is

$$z = f(x, y)$$

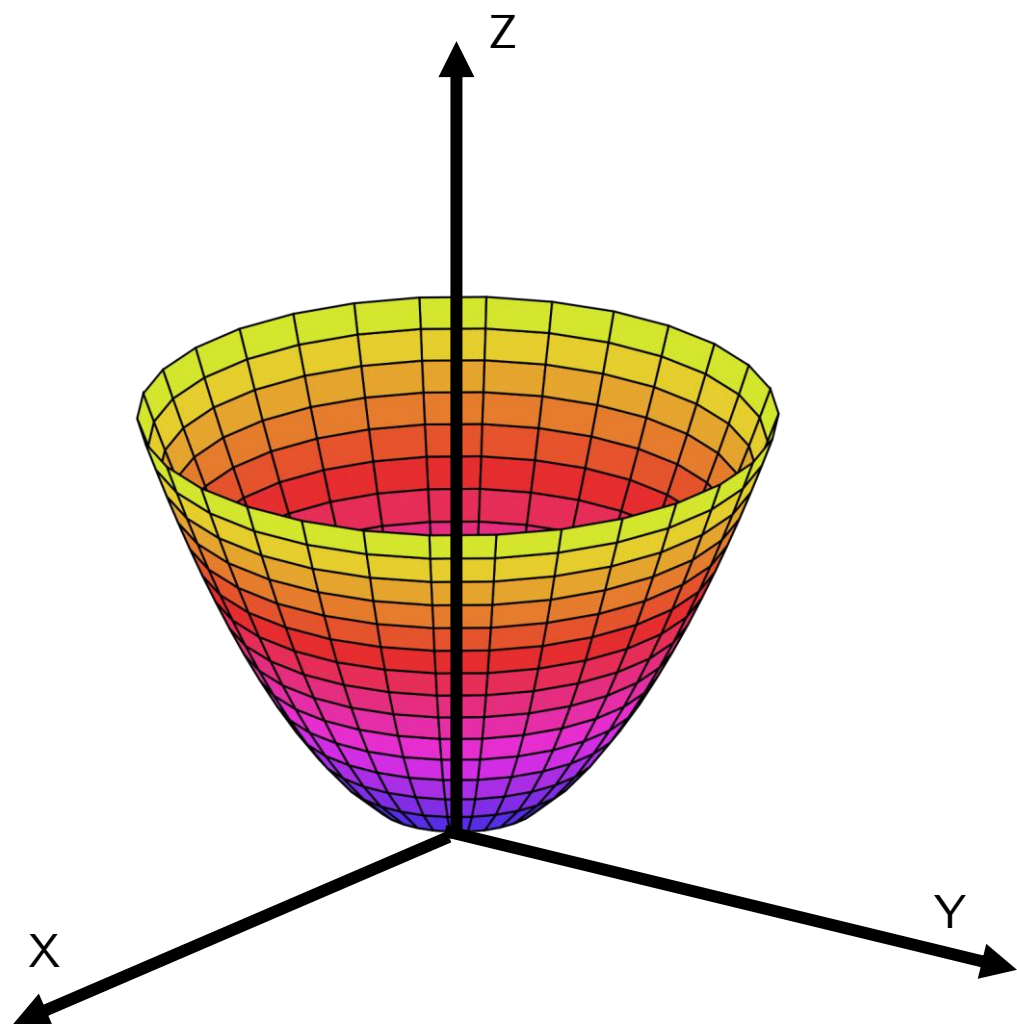
- Other examples include

Utility function : $u(x, y) = x^{\frac{1}{2}}y^{\frac{1}{2}}$

Option Price : $V(S, K, r, \tau, \sigma) = SN(d_1) - Ke^{-r\tau}N(d_2)$, $d_1 = \frac{\ln\left(\frac{S}{K}\right) + (r + 0.5\sigma^2)T}{\sigma\sqrt{T}}$, $d_2 = d_1 - \sigma\sqrt{T}$

- In this chapter, we will talk about partial derivatives which are used for
- I. Measuring the rate of change of output to input i.e., sensitivity or Greeks in finance,
 - II. Approximation of change in a function (multivariate Taylor Series) ,
 - III. Optimization (Maxima, Minima)

Visualizing in 3D ...



$$z = f(x, y) = x^2 + y^2$$

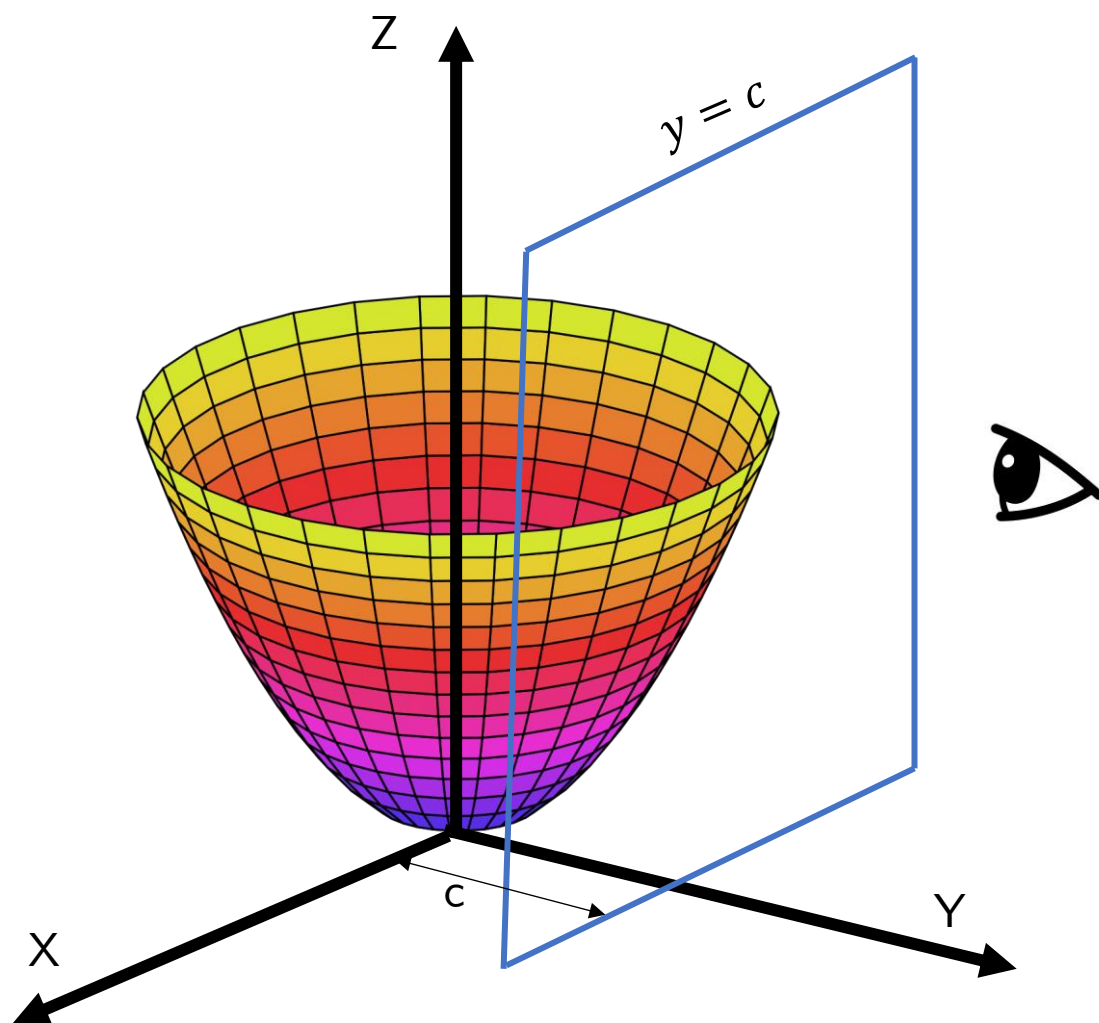
Output/Dependent
variable

Input/Independent
variables

Domain : $(x, y) \in \mathbb{R}^2$

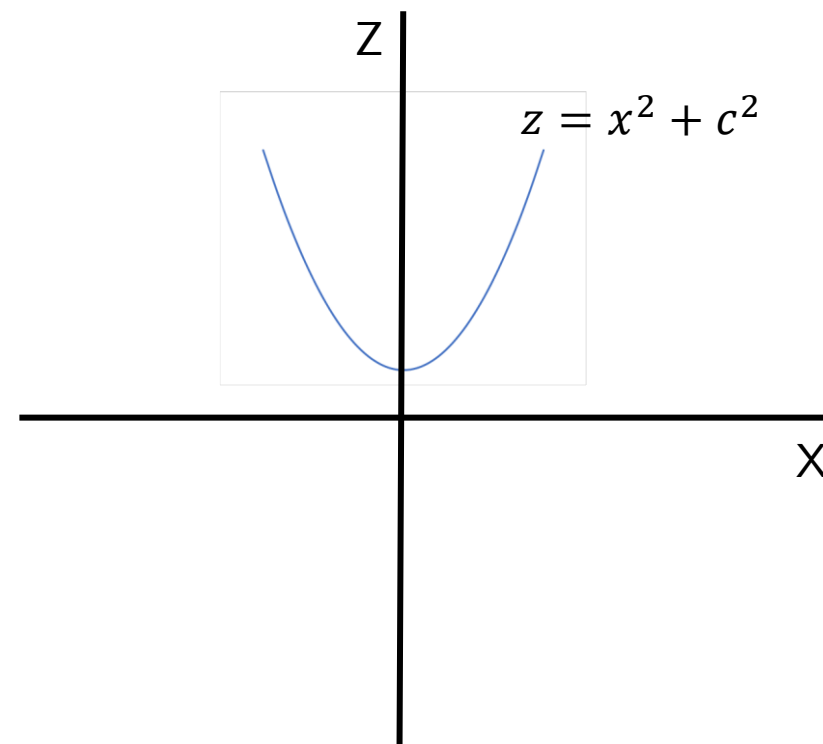
Range : $z \in \mathbb{R}^+$

Intersection of a plane and a surface

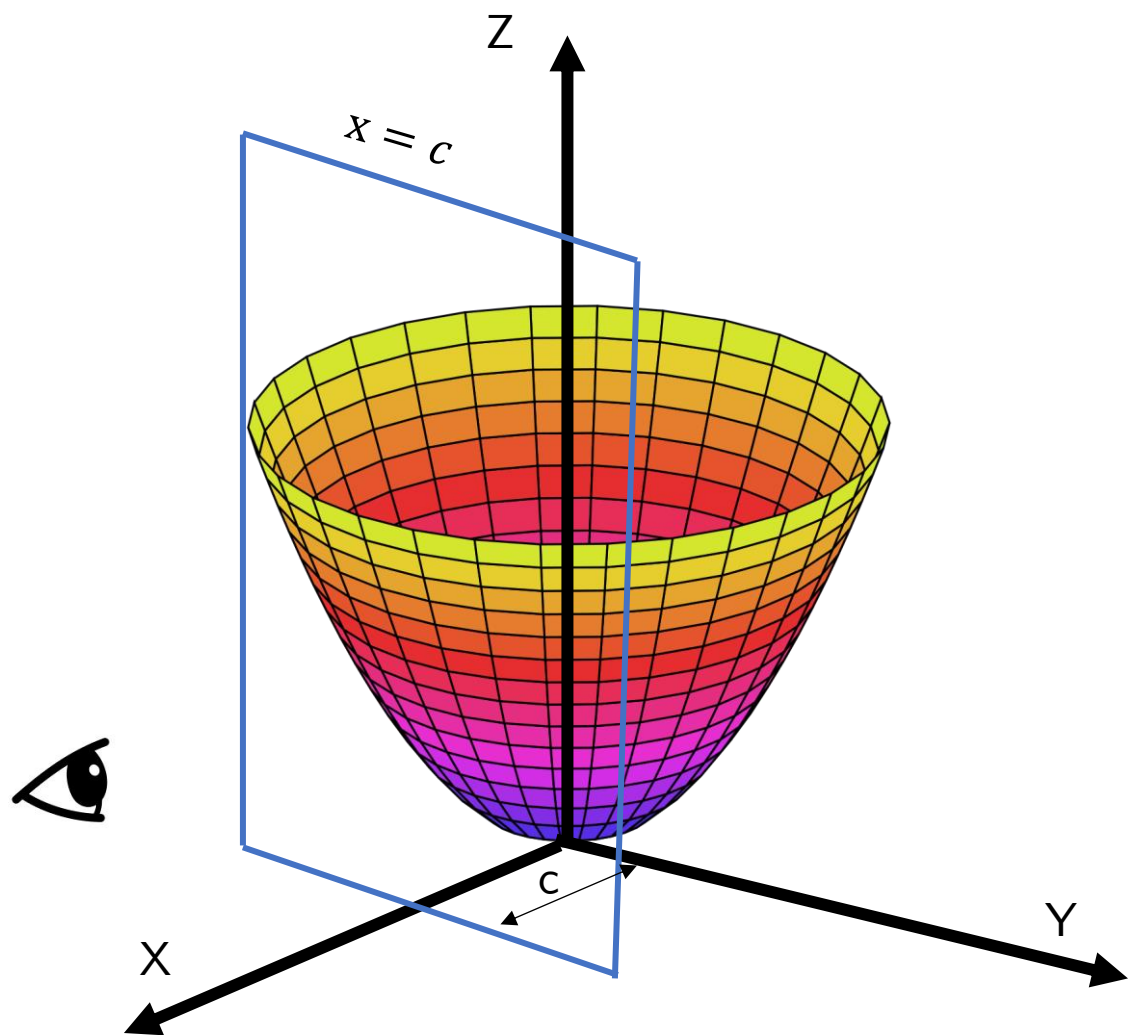


$$z = f(x, y) = x^2 + y^2$$

What is the shape of the section cut by the plane $y = c$?

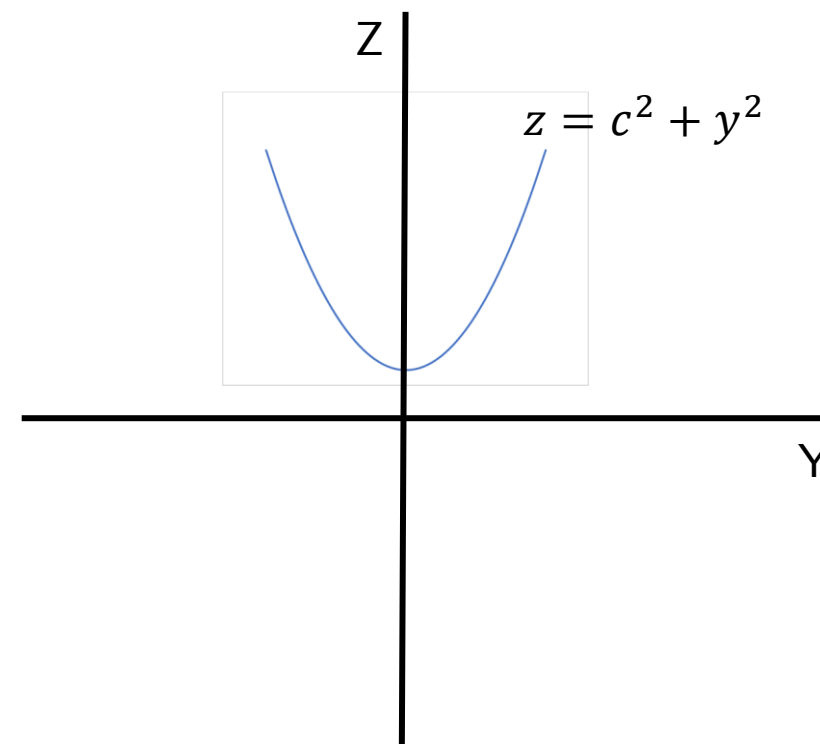


Intersection of a plane and a surface

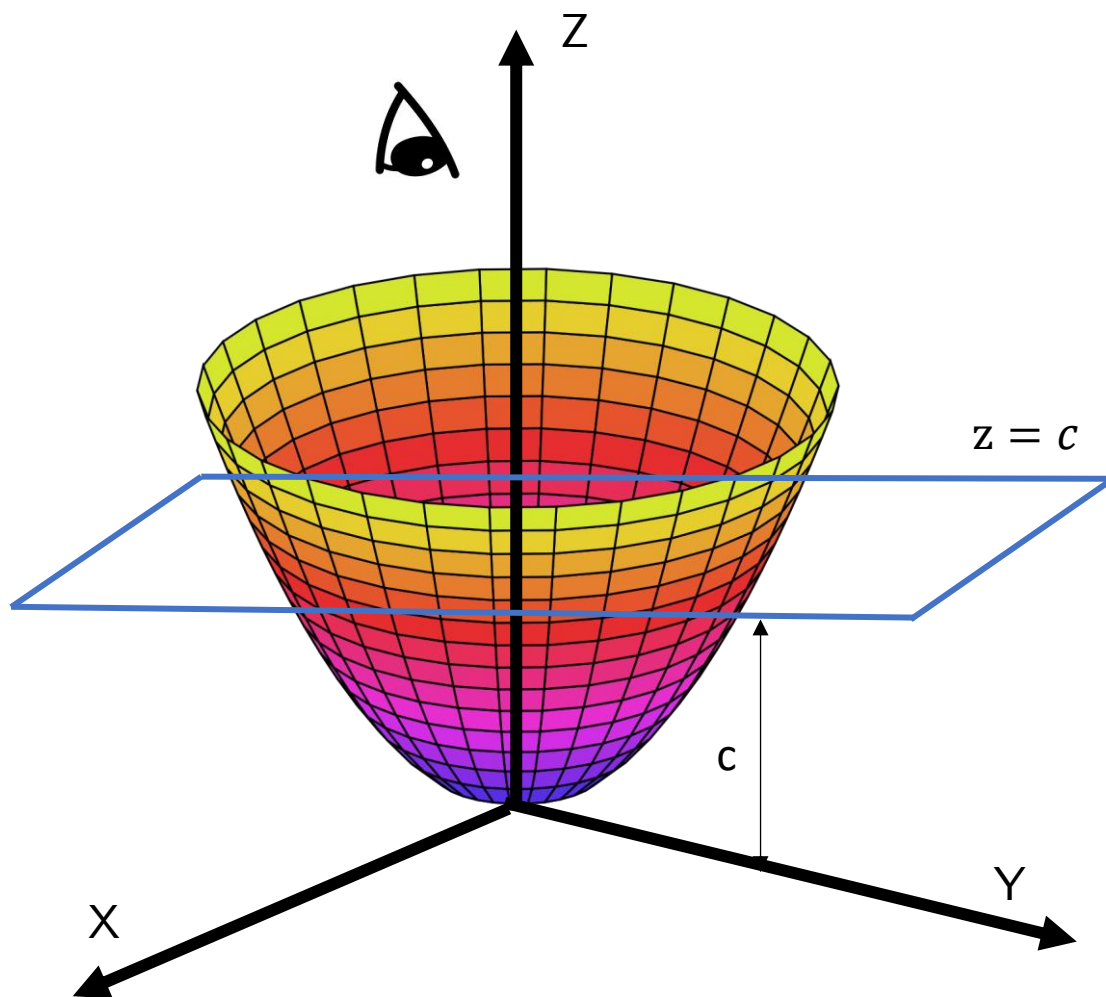


$$z = f(x, y) = x^2 + y^2$$

What is the shape of the section cut by the plane $x = c$?

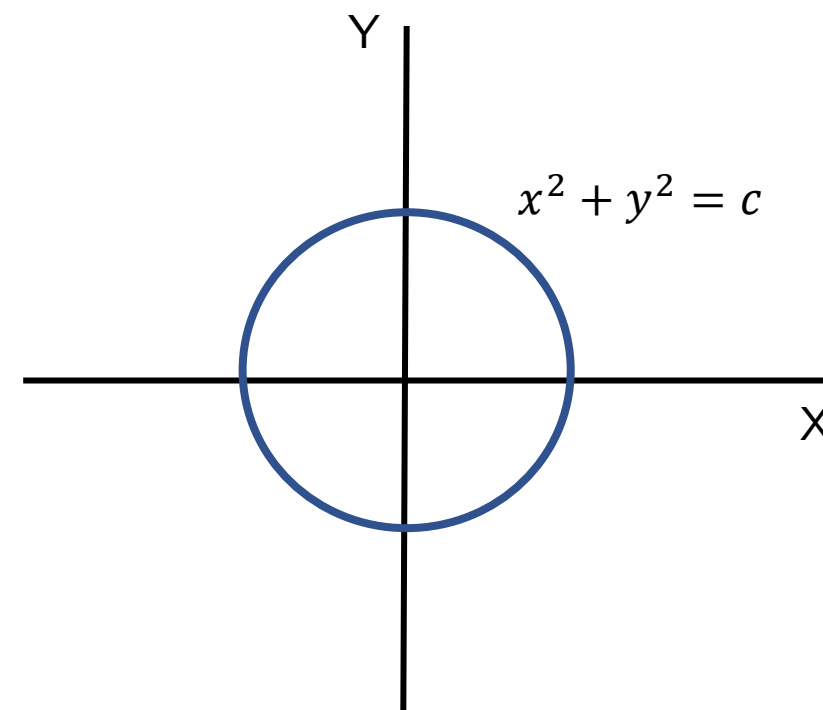


Level Curve

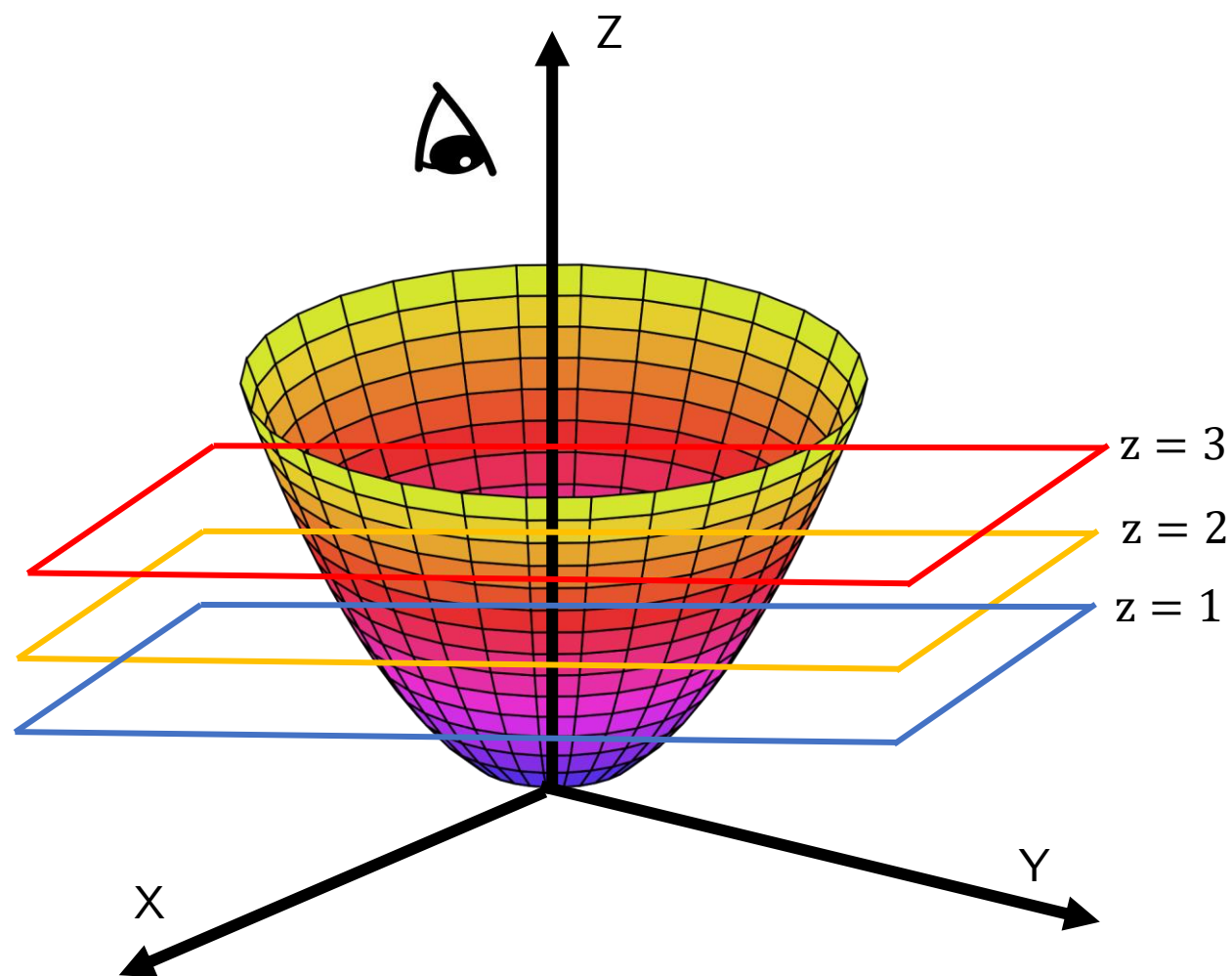


$$z = f(x, y) = x^2 + y^2$$

- ▶ If we fix $z = c$, we get a circle of points which share the same height $= c$.
- ▶ This is called as a **level curve**

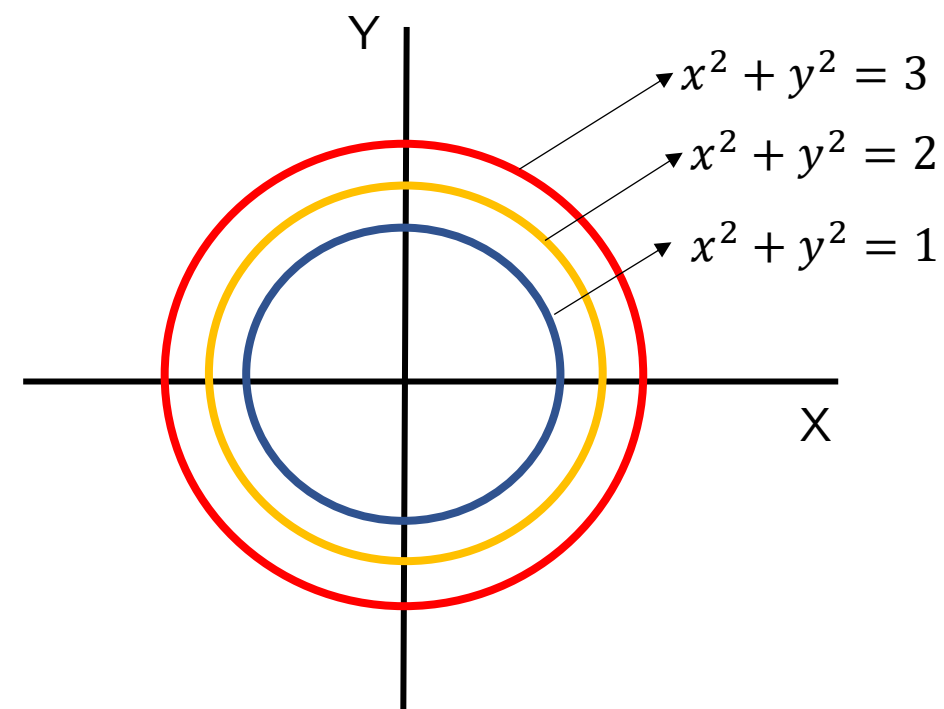


Contour Plot



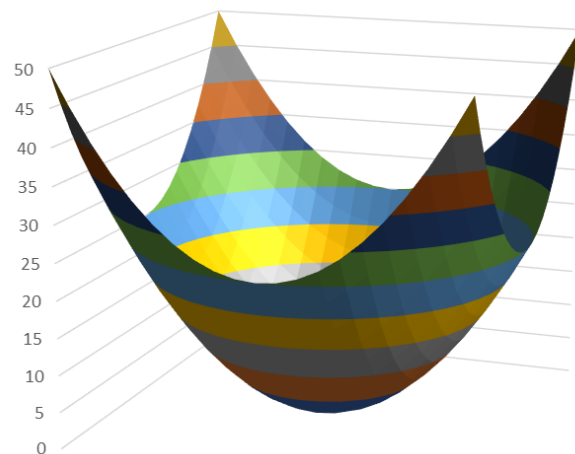
$$z = f(x, y) = x^2 + y^2$$

- By changing c , we can generate many level curves. The set is called **contour plot**.

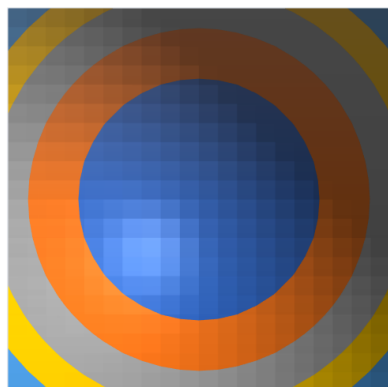


Contour Plots - Examples

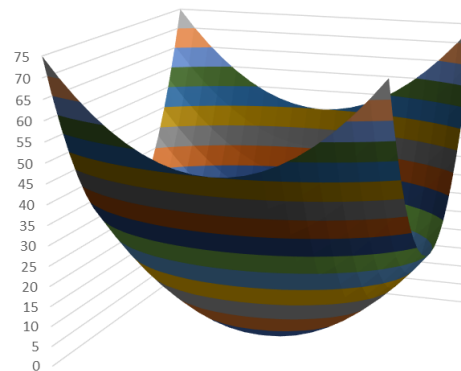
$$z = f(x, y) = x^2 + y^2$$



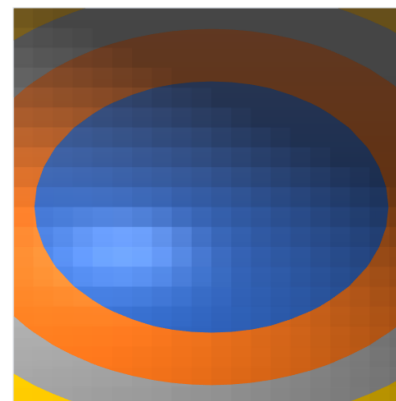
Contour lines are circles



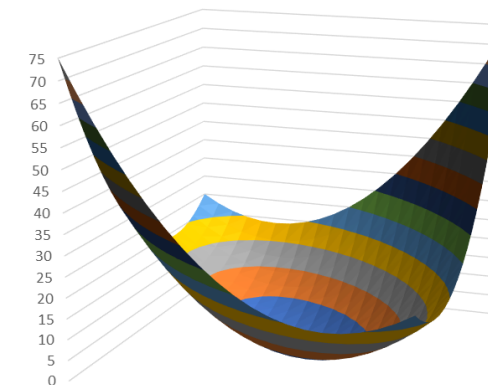
$$z = f(x, y) = x^2 + 2y^2$$



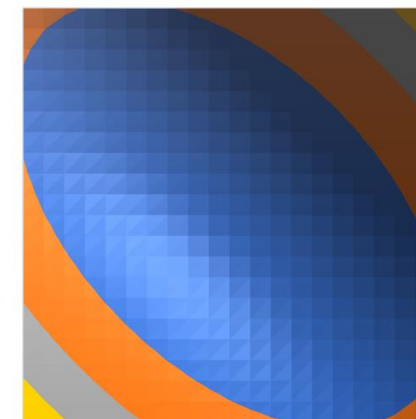
Contour lines are ellipses



$$z = f(x, y) = x^2 + xy + y^2$$



Contour lines are rotated ellipses

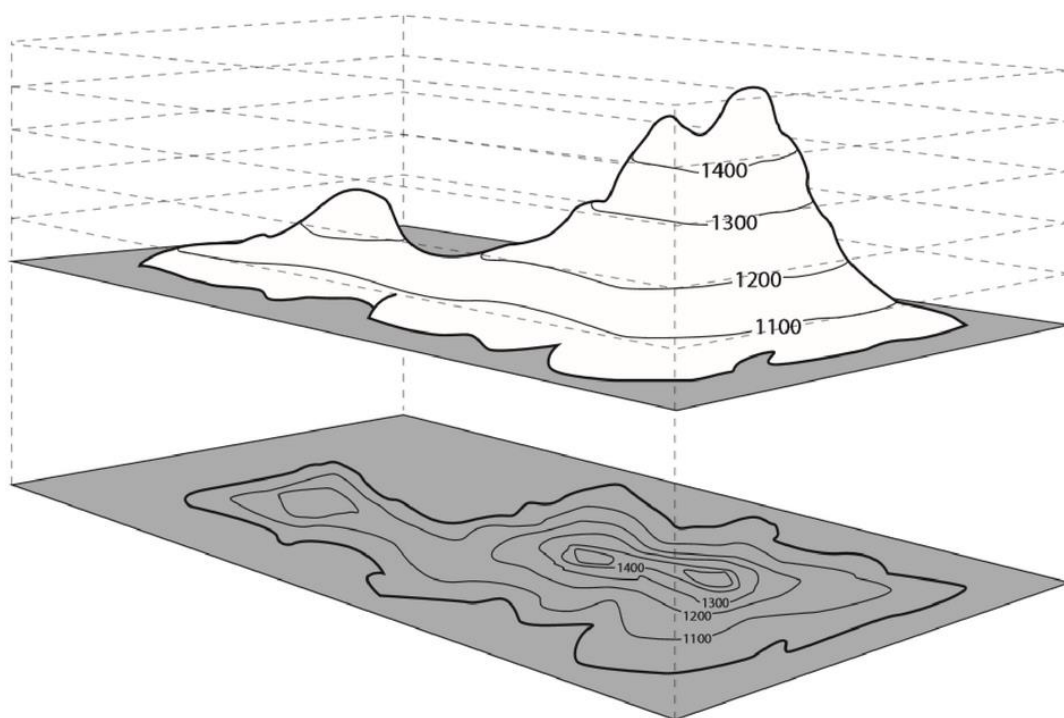


Topography and Heat Maps

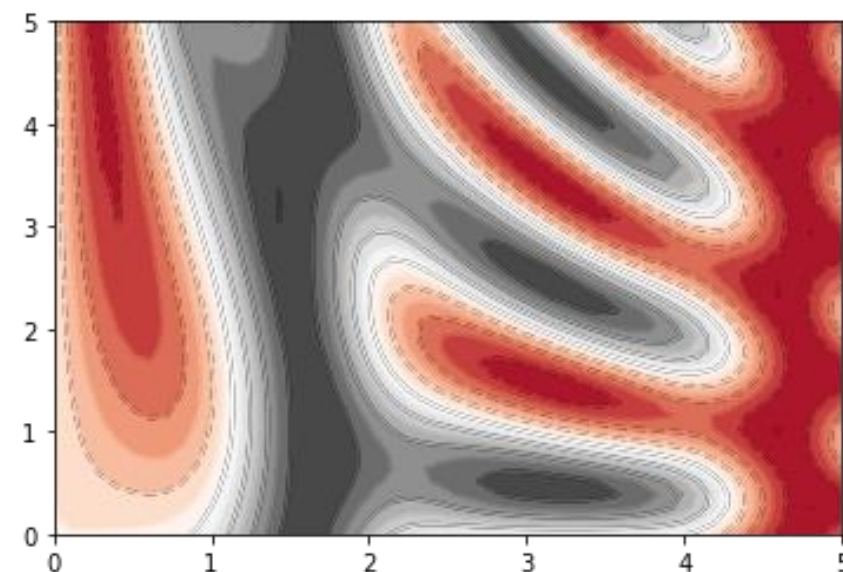


- ▶ Topographical plots and heatmaps are common visualization tools to gauge the elevation

Topographical map

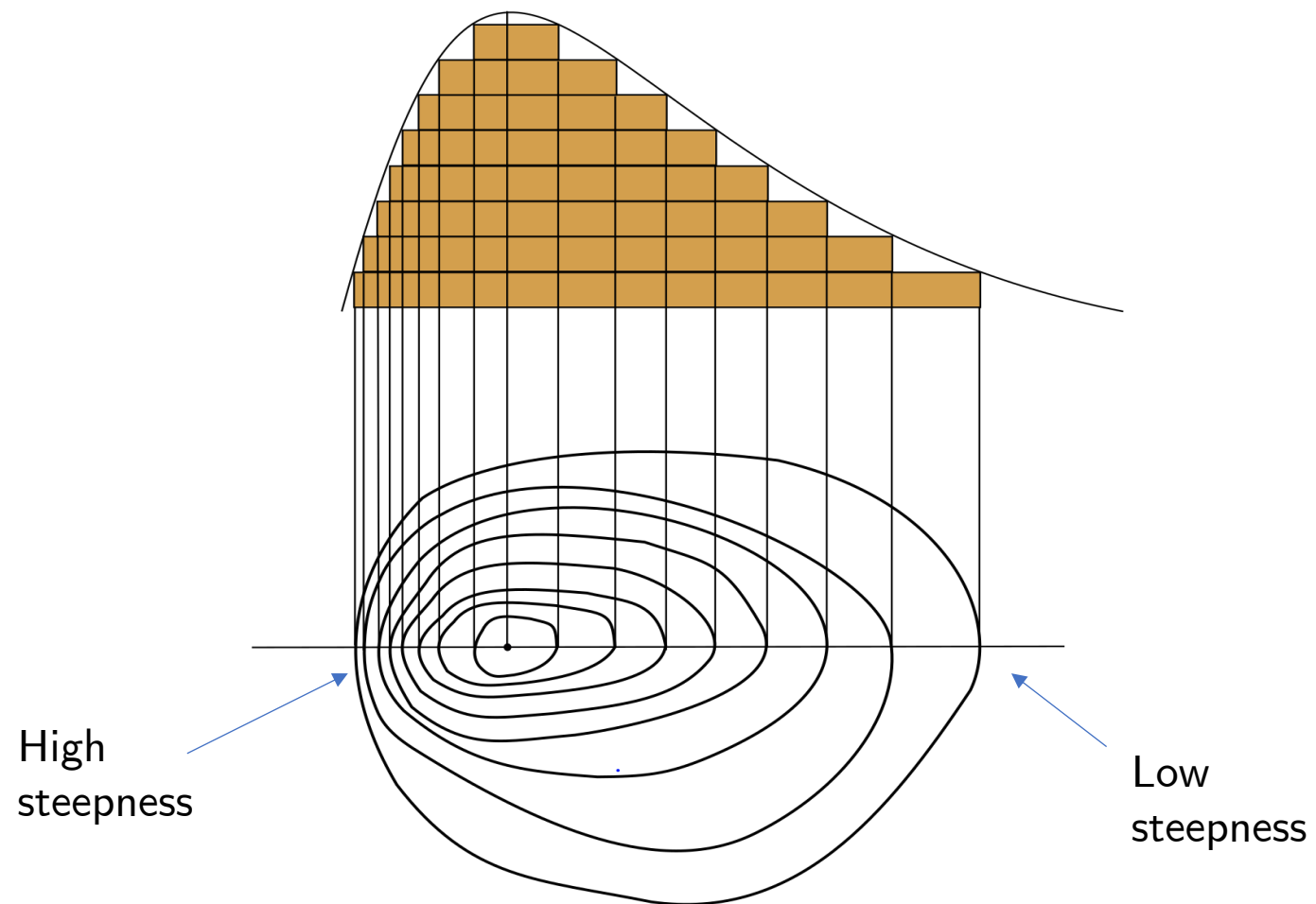


Heat Maps



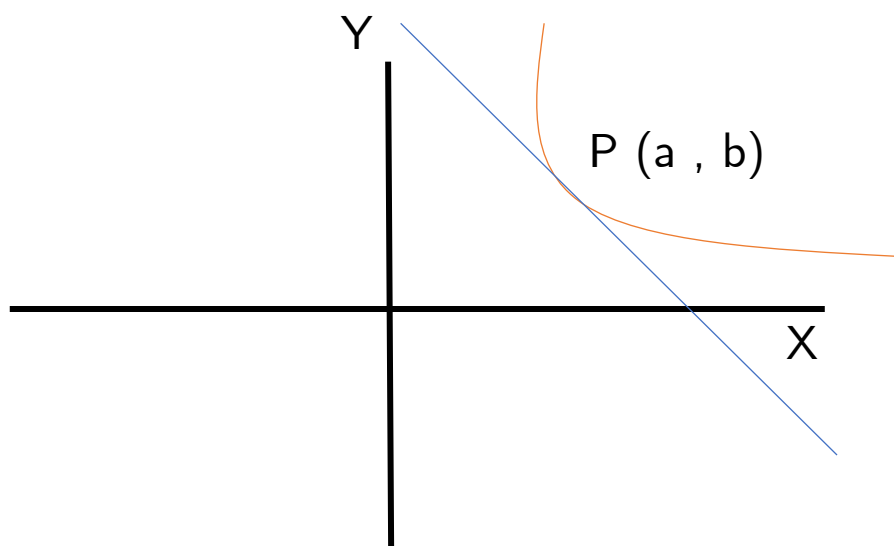
What does contour plot tell us ?

- ▶ Contour plots can give an idea of steepness.

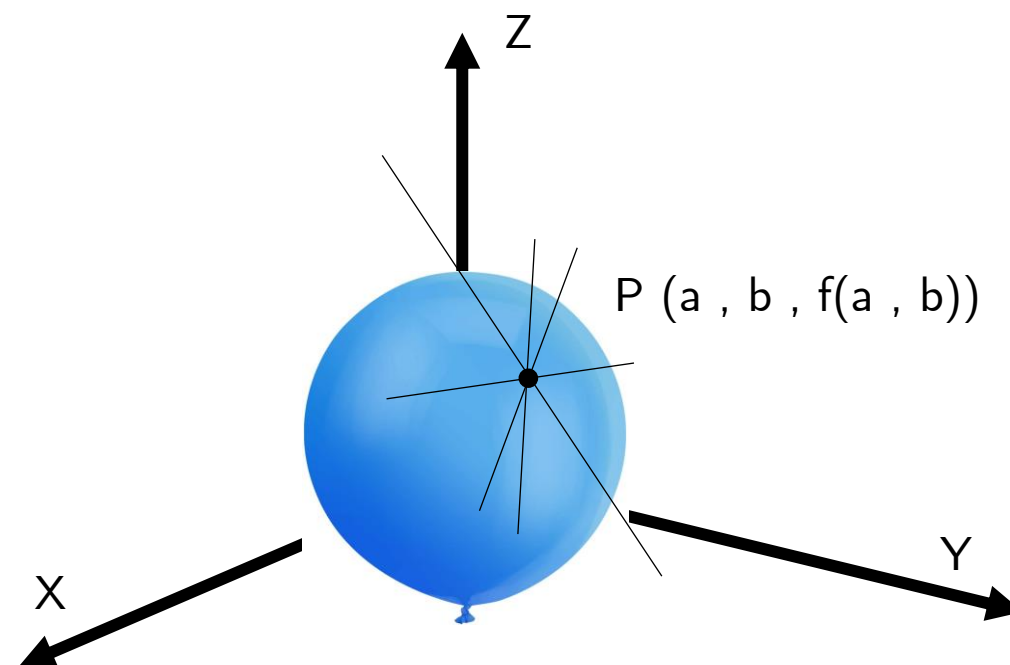


Derivatives in the presence of multiple variables

- For a curve in 2D, the derivative at a point is the slope of the tangent to the curve which is unique. But in case of multiple variables, the derivative at a point can exist in infinite directions (infinite tangents exist). Therefore, we need to specify a direction when we talk about differentiating in the presence of multiple input variables.



Derivative is the slope of the tangent to the curve

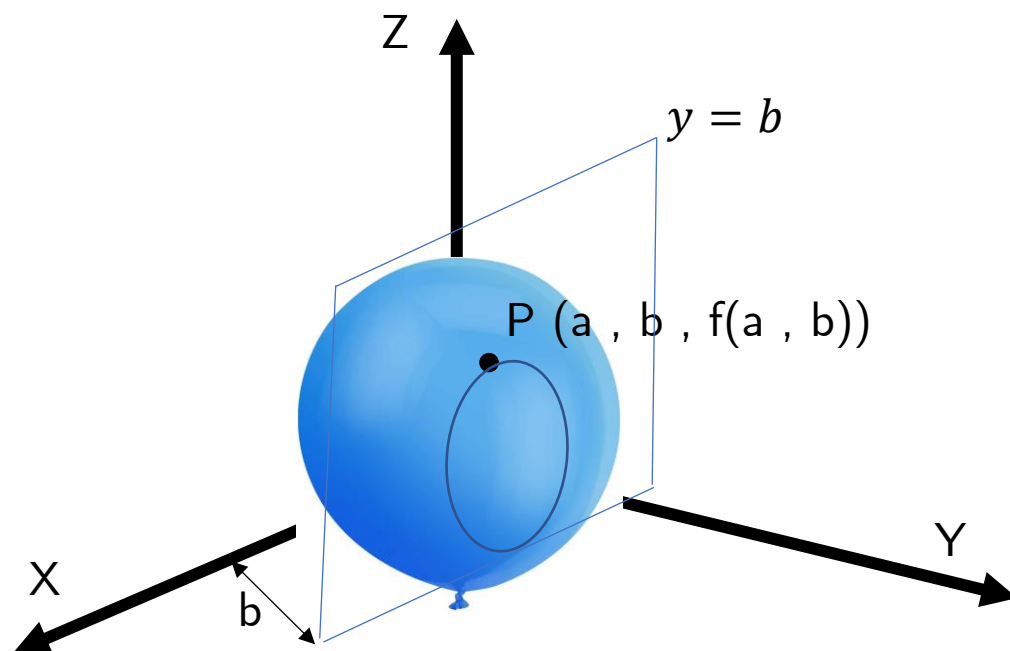


So many possible tangents !!

Derivative along $x := f_x$

- Consider a bivariate function $z = f(x, y)$. We can calculate the derivative of z w.r.t x without changing y . It's called as **partial derivative** along x and is denoted as

$$f_x = \frac{\partial f}{\partial x} \text{ (first order partial derivative w.r.t } x \text{)}$$



Derivative of z along x is ratio of change of z when x changes by a small amount keeping y constant

The curve generated by intersection of the surface and the plane $y = b$, x is the only independent variable, and y is kept constant at b

$z = f(x, y)$ → surface

$z = f(x, b)$ → curve

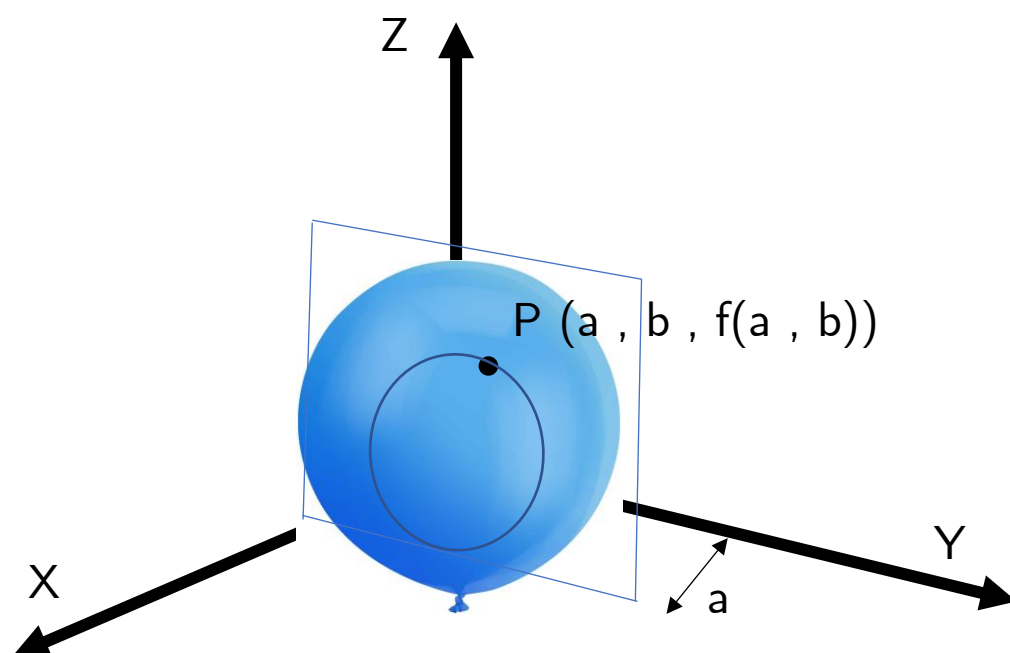
$$\frac{\partial f}{\partial x} = f_x = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

$$\left(\frac{\partial f}{\partial x} \right)_{x=a, y=b} = \lim_{\Delta x \rightarrow 0} \frac{f(a + \Delta x, b) - f(a, b)}{\Delta x}$$

Derivative along $y := f_y$

- We can similarly calculate partial derivative along y

$$f_y = \frac{\partial f}{\partial y} \text{ (first order partial derivative w.r.t } y \text{)}$$



Derivative of z along y is ratio of change of z when y changes by a small amount keeping x constant

The curve generated by intersection of the surface and the plane $x = a$, y is the only independent variable, and x is kept constant at a

$$z = f(x, y) \longrightarrow \text{surface}$$

$$z = f(a, y) \longrightarrow \text{curve}$$

$$\frac{\partial f}{\partial y} = f_y = \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$$

$$\left(\frac{\partial f}{\partial y} \right)_{x=a, y=b} = \lim_{\Delta y \rightarrow 0} \frac{f(a, b + \Delta y) - f(a, b)}{\Delta y}$$



First-order Partial derivatives

Exercises : Compute the first-order partial derivatives for the following functions

1. $f(x, y) = 10 + 2x + 3y$

2. $f(x, y) = 10 + 2x^2 + y^3$

3. $f(x, y) = 3x^2y^2$

4. $f(x, y, z) = 10x + 12y + 20z$

5. $f(x, y) = 2x^2(4x + 6y)$

6. $f(x, y) = \frac{x+y}{2y}$



Second-order Partial derivatives

► By taking partial derivatives of partial derivatives, we get higher order partials. For example, the following are defined

$$\text{Second order partials : } f_{xx} = \frac{\partial f_x}{\partial x} = \frac{\partial^2 f}{\partial x^2}, f_{yy} = \frac{\partial f_y}{\partial y} = \frac{\partial^2 f}{\partial y^2}$$

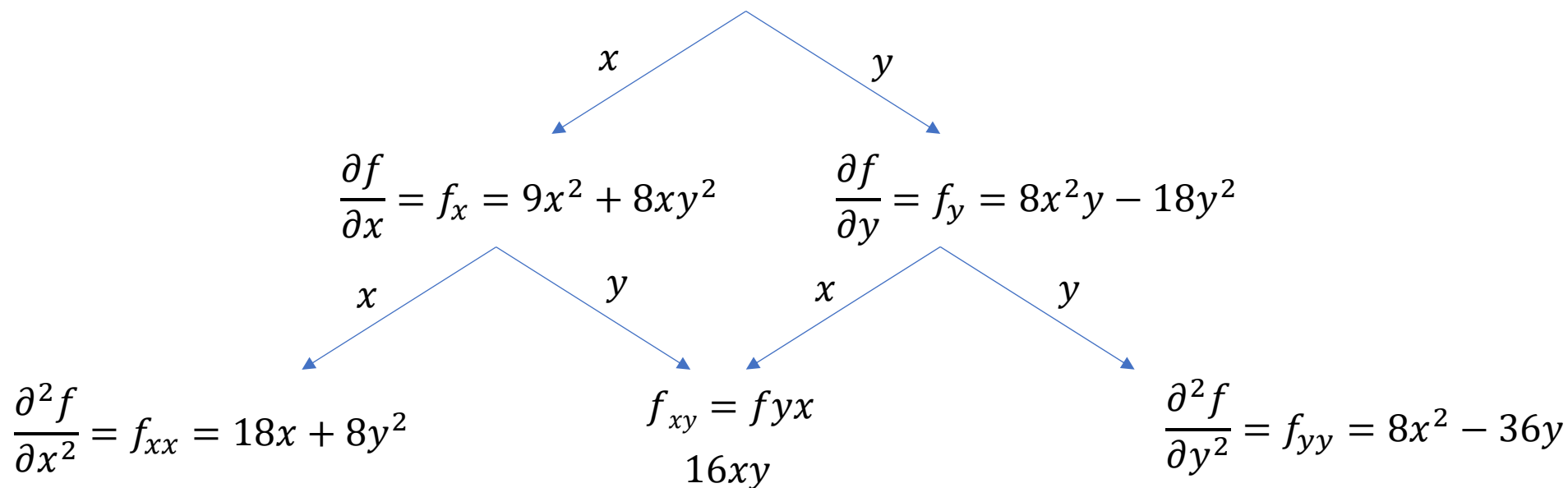
$$\text{Mixed partials : } f_{xy} = \frac{\partial f_x}{\partial y}, f_{yx} = \frac{\partial f_y}{\partial x}$$

$$\frac{\partial f_x}{\partial y} = \frac{\partial f_y}{\partial x} \quad (\text{Young's theorem})$$

Exercise : Show that $f_{xy} = f_{yx}$ for $f(x, y) = 3x^3 + 4x^2y^2 - 6y^3$

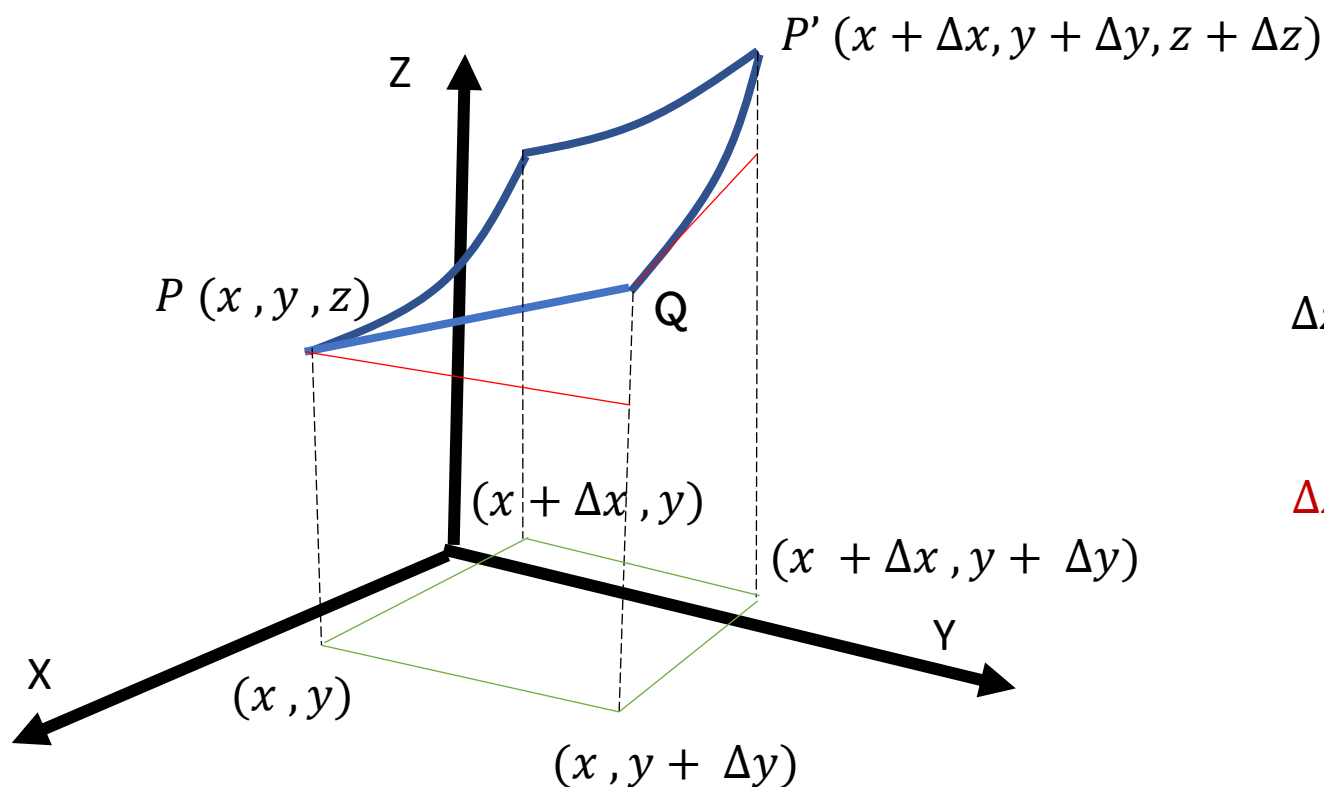
Second-order Partial derivatives

$$z = f(x, y) = 3x^3 + 4x^2y^2 - 6y^3$$



Total Derivative

- Total differential of a function can be attributed to changes in the inputs through the partial derivatives. We can visualize how much is because of first order change



$$\Delta z = \begin{array}{l} \text{(change because of change in } y) \\ + \\ \text{(change because of change in } x) \end{array}$$

$$\Delta z = z_y \Delta y + z_x \Delta x$$

Multivariate Taylor Series



- Now that we have functions of multiple inputs, the Taylor series approximation for the change in the function can be defined up to second order as

$$\Delta f(x, y) \approx f_y \Delta y + f_x \Delta x + \frac{1}{2} f_{yy} (\Delta y)^2 + \frac{1}{2} f_{xx} (\Delta x)^2 + f_{xy} (\Delta x) (\Delta y)$$

- Notice there is no $\frac{1}{2}$ in the mixed partial term because it represents the sum of 2 mixed partials which are same

Exercise : Express the total change in $f(x, y) = e^{x+y}$

Chain Rule



- In case of univariate functions, the chain rule we learnt was straightforward. For example, if $y = x^2$ and $x = t^3$, then

$$\frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt} = 2x \cdot 3t^2 = 6t^5$$

- In case of functions with several variables, the total derivative is a bit involved. For example, find the following

$$z = f(x, y) = x^2 + y^2$$

and $y = t^2$ and $x = t^3$, find $\frac{dz}{dt}$

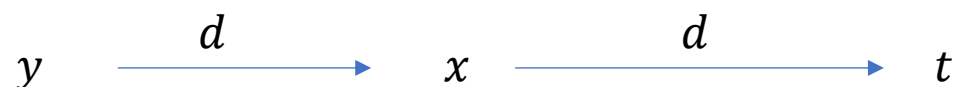
- Using $dz = f_x dx + f_y dy \Rightarrow \frac{dz}{dt} = f_x \frac{dx}{dt} + f_y \frac{dy}{dt} = 2x \cdot 3t^2 + 2y \cdot 2t = 6t^5 + 4t^3$

Chain Rule

- So basically, with multiple input variables, we need to include changes from all possible paths.
Let's look at some examples

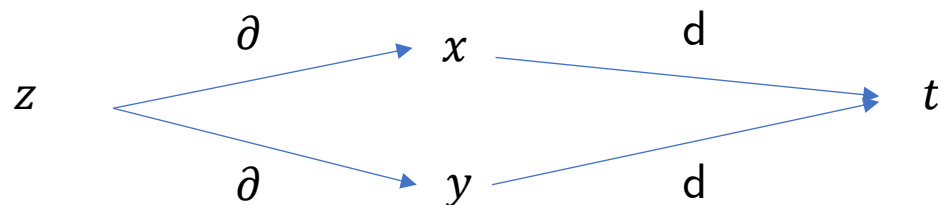
$$y = x^2 \text{ and } x = t^3$$

dependent Intermediate Independent



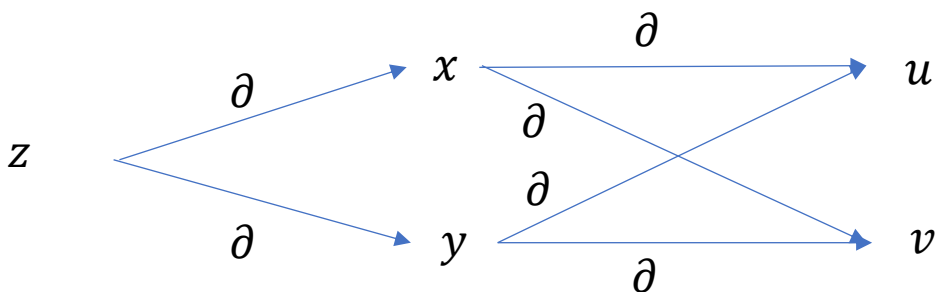
$$z = f(x, y) = x^2 + y^2$$

$$y = t^2 \text{ and } x = t^3$$



$$z = x^3 + y^3$$

$$x = u^2 + v^2 \quad y = 2uv$$





Gradient of a scalar function

- Consider a scalar valued function maps to a real number $f: \mathbb{R}^n \rightarrow \mathbb{R}$. When we represent the partial derivatives along x and y in vector form, it's called the **gradient vector** $\nabla z: \mathbb{R}^n \rightarrow \mathbb{R}^n$

$$z = f(x, y) \quad f: \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$\nabla z = \left(\frac{\partial f}{\partial x} \quad \frac{\partial f}{\partial y} \right) \quad (\text{gradient vector})$$

- Gradient of a scalar function $z = f(x, y)$ produces a vector ∇z

Exercise : Compute the gradient of $w = (x^2 + y^2)$ and show that the gradient at $(x_0 = 1, y_0 = 2)$, is perpendicular to the level curve at that point.



Jacobian (Gradient of a vector function)

▶ A vector valued function maps to a vector $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$

$$w = f(x, y) = \begin{pmatrix} f_1(x, y) \\ f_2(x, y) \end{pmatrix} = \begin{pmatrix} 2xy \\ x \ln y \end{pmatrix}$$

(vector valued function)

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

The Jacobian $\mathbf{J}$ of $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ will be a $(m \times n)$ matrix where $\mathbf{J}_{i,j} = \frac{\partial f_i}{\partial x_j}$. (i^{th} output w.r.t j^{th} input).

In the example above, Jacobian is given as

$$\mathbf{J}_f = \begin{pmatrix} \nabla f_1 \\ \nabla f_2 \end{pmatrix} = \begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{pmatrix} = \begin{pmatrix} 2y & 2x \\ \ln y & x/y \end{pmatrix}$$

Hessian



- The **Hessian Matrix** is a square matrix of 2nd order partial derivatives of a scalar valued function. It's given by the *Jacobian of the transpose of gradient*.

$$(\nabla_Z)^T = \begin{pmatrix} f_x \\ f_y \end{pmatrix}$$

$$\mathcal{H}_f = \mathcal{J}_{\nabla_Z^T} = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} \quad (\text{Hessian matrix})$$

- For example, if $w = f(x, y) = x^2y$, then

$$\nabla w = (2xy \quad x^2)$$

$$\text{and } \mathcal{H}_f = \begin{pmatrix} 2y & 2x \\ 2x & 0 \end{pmatrix}$$

Quadratic Approximation of a Multivariable Function



- The Taylor series approximation till 2nd order (Quadratic Approximation) for a scalar valued function can be written in a more compact way

$$f(\mathbf{x}) = \underbrace{f(\mathbf{x}_0)}_{\text{(Const.)}} + \underbrace{\nabla f(\mathbf{x}_0) \cdot (\mathbf{x} - \mathbf{x}_0)}_{\text{(Linear)}} + \underbrace{\frac{1}{2} (\mathbf{x} - \mathbf{x}_0)^T \mathcal{H}_f(\mathbf{x}_0) (\mathbf{x} - \mathbf{x}_0)}_{\text{(Quadratic)}}$$

Exercise – Find the quadratic approximation of $f(x, y) = xe^y$ at $\mathbf{x}_0 = (0 \ 0)$

Exercises



1. if $U(x, y, z) = 2x^2 - yz + xz^2$, where $x = 2 \sin t$, $y = t^2 - t + 1$, $z = 3e^{-t}$, show that

$$\frac{dU}{dt} \text{ at } t = 0 \text{ is } 24.$$

2. Given $w = f(x, y)$; $x = r \cos \theta$, $y = r \sin \theta$, show that

$$\left(\frac{\partial w}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial w}{\partial \theta}\right)^2 = \left(\frac{\partial w}{\partial x}\right)^2 + \left(\frac{\partial w}{\partial y}\right)^2$$